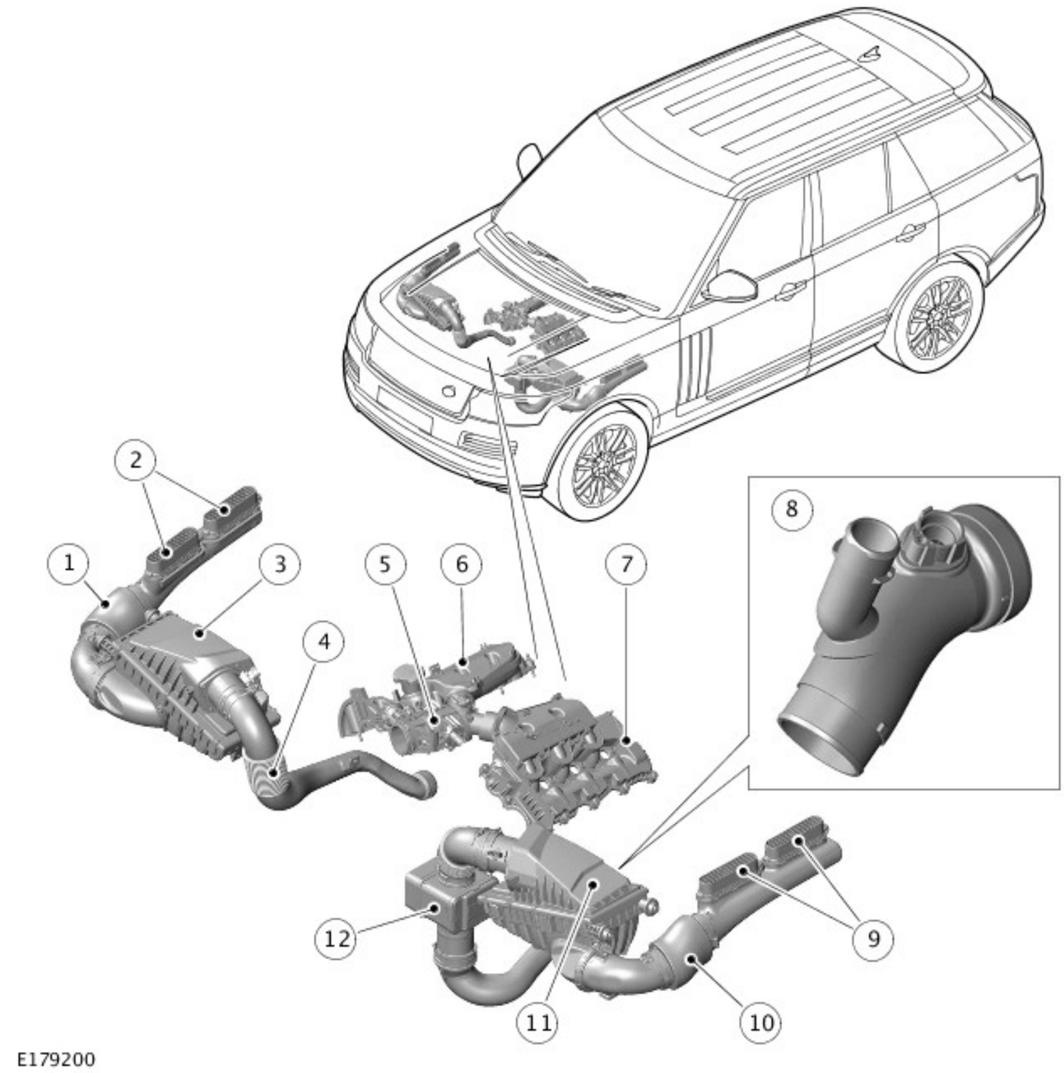
2016.0 RANGE ROVER (LG), 303-12

INTAKE AIR DISTRIBUTION AND FILTERING - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

DESCRIPTION AND OPERATION

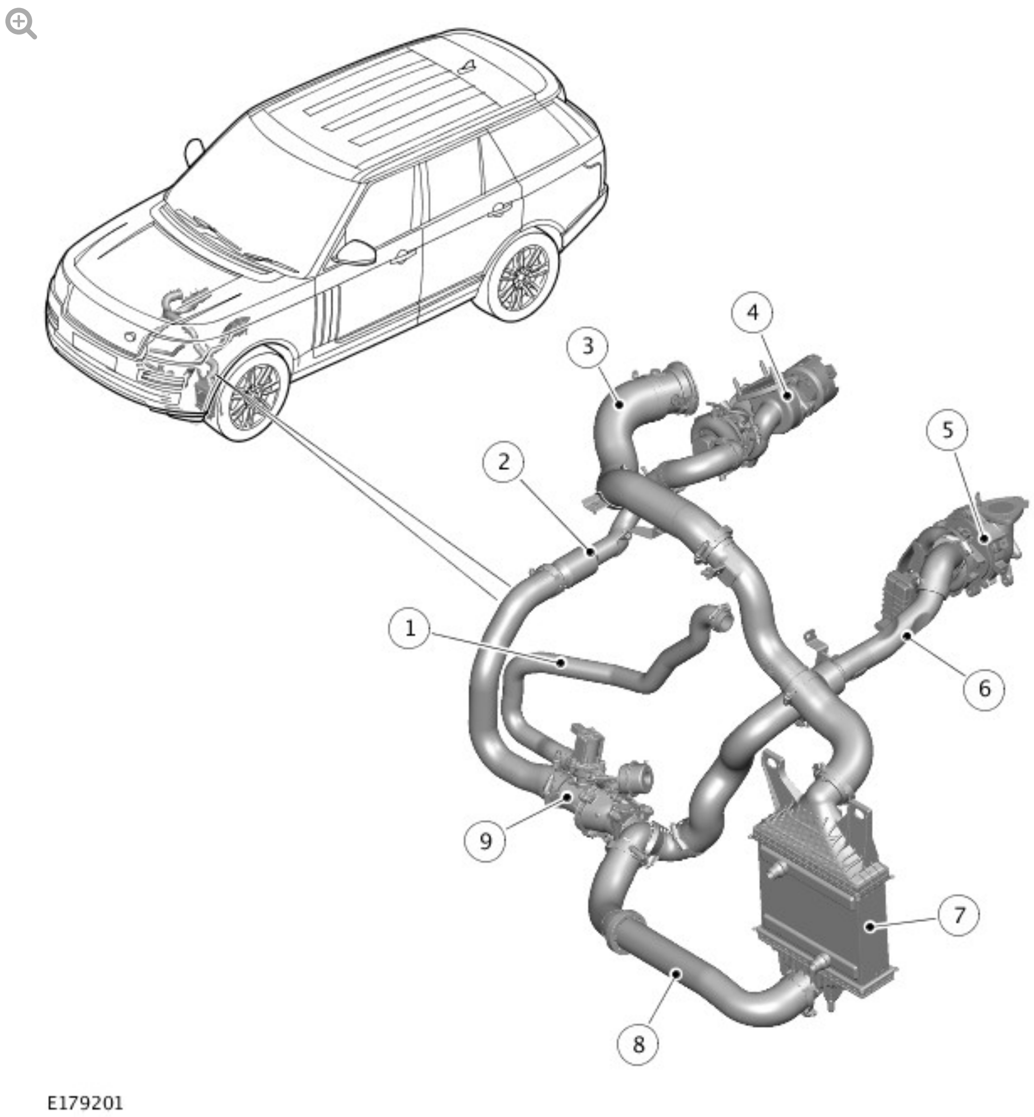
COMPONENT LOCATION - SHEET 1 OF 3



ITEM	DESCRIPTION
1	Right dirty air duct (if fitted)
2	Right intake grilles (if fitted)
3	Right air cleaner (if fitted)
4	Right clean air ducts
5	Electric throttle
6	Right intake manifold (integral with camshaft cover)
7	Left intake manifold (integral with camshaft cover)
8	EGR pipe (if fitted)
9	Left intake grilles
10	Left dirty air duct

11	Left air cleaner
12	Left clean air ducts with resonator

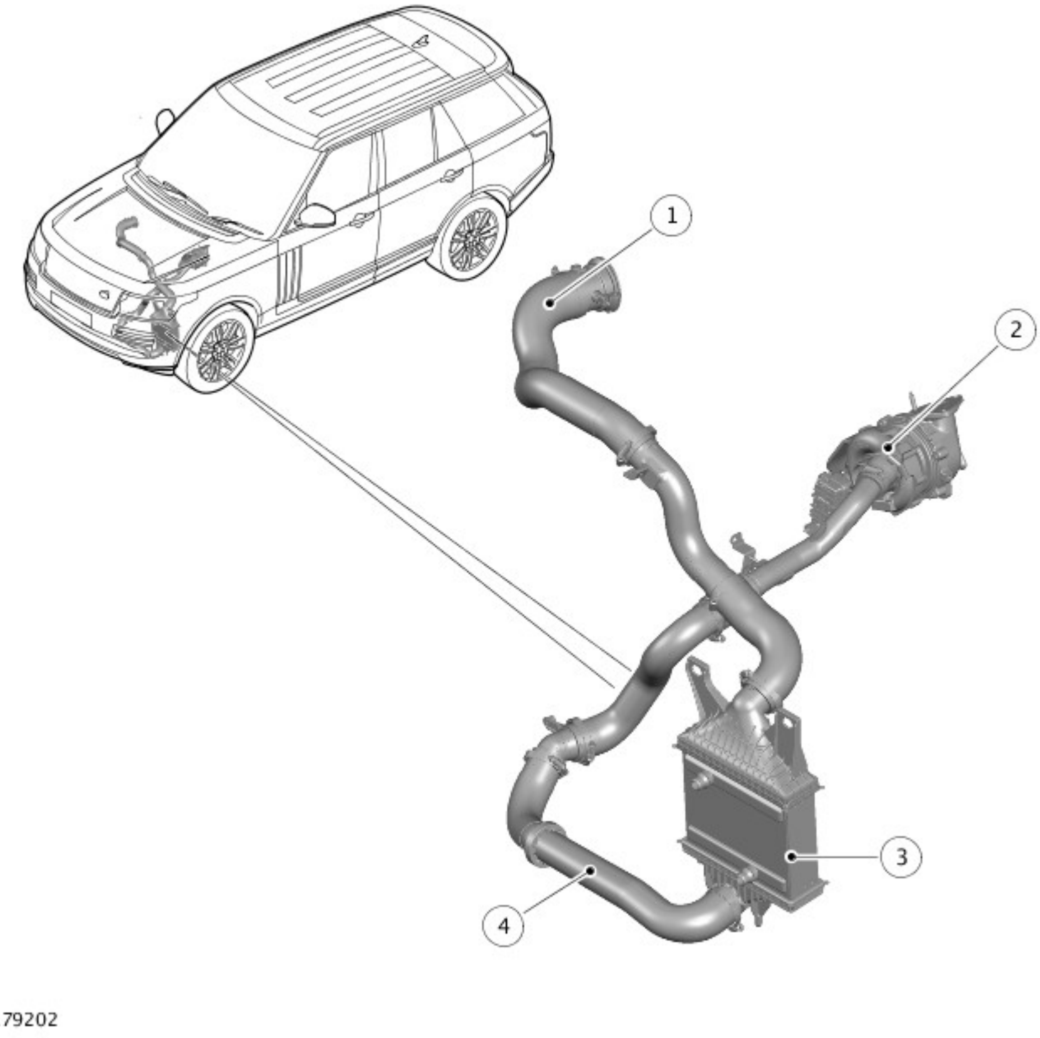
COMPONENT LOCATION - SHEET 2 OF 3 - TWIN-TURBO VARIANT



ITEM	DESCRIPTION
1	Charge air duct – charge air intake valve to left clean air duct
2	Charge air ducts – secondary turbocharger to charge air intake valve
3	Charge air ducts – charge air cooler to electric throttle
4	Secondary turbocharger

5	Primary turbocharger
6	Charge air ducts – primary turbocharger to charge air cooler
7	Charge air cooler
8	Charge air ducts – charge air intake valve to charger air cooler
9	Charge air intake valve

COMPONENT LOCATION – SHEET 3 OF 3 – MONO-TURBO VARIANT



ITEM	DESCRIPTION
1	Charge air ducts – charge air cooler to electric throttle
2	Turbocharger

3	Charge air cooler
4	Charge air ducts – turbocharger to charge air cooler

OVERVIEW

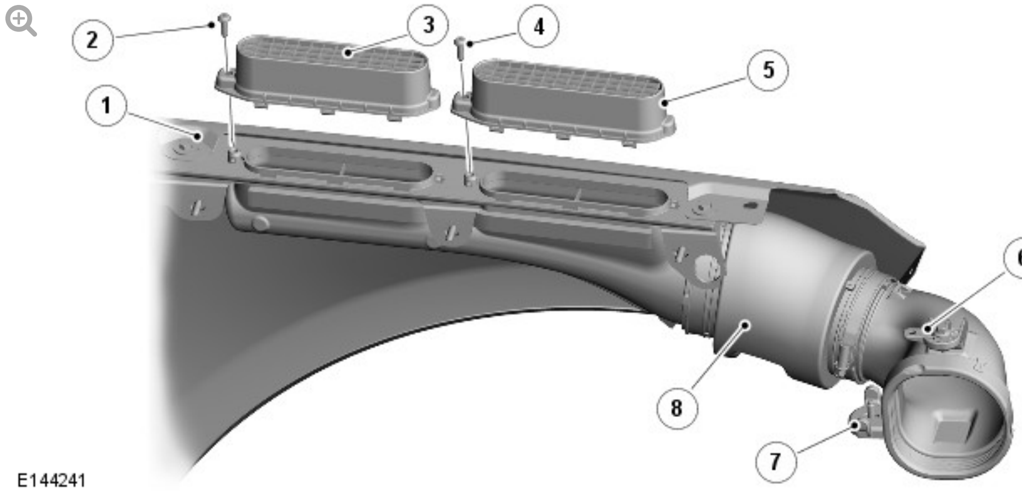
The intake air distribution and filtering system cleans, compresses and cools the intake air. The system comprises according to bi- or mono-turbo variant is equipped:

Bi-turbo engines	**Mono-turbo engines**
Two dirty air ducts.	A dirty air duct.
Two air cleaners.	An air cleaner.
Two clean air ducts, one with resonator.	A clean air duct with resonator.
Two turbochargers.	A turbocharger.
Charge air ducts.	A charge air duct.
A charge air intake valve.	-
A water cooled charge air cooler.	A water cooled charge air cooler.
An electric throttle.	An electric throttle.
Two intake manifolds.	Two intake manifolds.
A low pressure EGR duct	A low pressure EGR duct

Air flows through the dirty air ducts, air cleaners and clean air ducts to the compressor intakes of the turbochargers. The turbochargers compress the air, which is then transferred through the charge air ducts to the charge air intake valve, charge air cooler, electric throttle and the intake manifolds.

DESCRIPTION

DIRTY AIR DUCTS



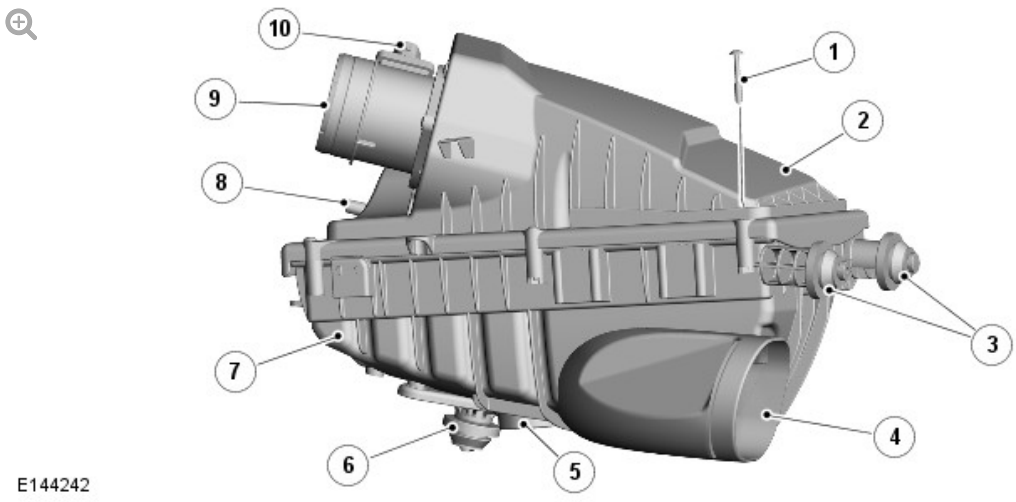
ITEM	DESCRIPTION
1	Front fender support
2	Screw
3	Intake grille
4	Screw
5	Intake grille
6	Front upper bracket
7	Front lower bracket
8	Non-porous sleeve

The dirty air ducts are installed in the front fenders to transfer ambient air from under the edges of the hood to the air cleaners. If mono-turbo variant is equipped one air duct is fitted on the left side.

Each dirty air duct locates in the intake of the related air cleaner and is attached to the front inner fender with two brackets. The intake of each dirty air duct is attached to two intake grilles on top of the front fender support. When the hood is closed, the intake grilles locate in guides attached to the hood inner panel.

A porous section, covered by a non-porous sleeve to prevent moisture ingress, is incorporated into each dirty air duct to reduce induction noise.

AIR CLEANERS



ITEM	DESCRIPTION
1	Screw (6 off)
2	Lid
3	Isolators
4	Air intake
5	Drain valve
6	Isolator
7	Base
8	Engine vacuum system connection (left air cleaner only)
9	Air outlet
10	Mass air flow sensor (left air cleaner) or mass air flow and temperature sensor (right air cleaner)

An air cleaner is located in each front corner of the engine compartment if bi-turbo variant is equipped and one air cleaner is located in the left front corner of the engine compartment if mono-turbo variant is equipped. Three spigots fitted with isolator bushes locate each air cleaner in holes in the related front inner fender and front suspension housing.

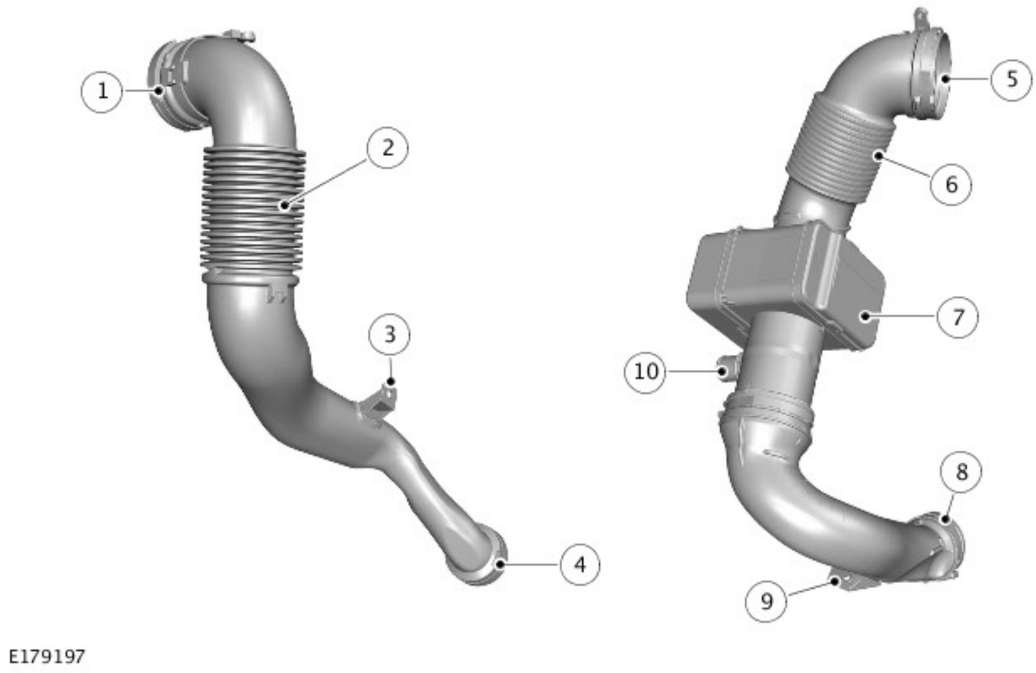
Each air cleaner consists of a filter element installed in a base and enclosed with a lid secured by six screws. The filter element is a pleated, paper type with an integral seal. Air intake and outlet connections are incorporated into the base and lid respectively. The bottom of the base incorporates a drain valve to prevent the accumulation of water in the air cleaner.

The lid of the left air cleaner incorporates a connection for the engine vacuum system. For additional information, refer to:

Turbocharger (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Description and Operation),
Engine Emission Control (303-08A Engine Emission Control - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).

The air outlet connection of each air cleaner incorporates either a mass air flow (MAF) sensor (right air cleaner, if equipped) and a mass air flow and temperature (MAFT) sensor (left air cleaner). If mono-tubro variant if equipped only a MAFT sensors is fitted. The sensors are connected to the Engine Control module (ECM).
For additional information, refer to: Electronic Engine Controls (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation).

CLEAN AIR DUCTS



ITEM	DESCRIPTION
1	Air intake - twin-turbo variant only
2	Convolute - twin-turbo variant only
3	Bracket - twin-turbo variant only
4	Air outlet - twin-turbo variant only
5	Air intake
6	Convolute
7	Resonator
8	Air outlet
9	Bracket
10	Charge air bypass hose connection

The clean air ducts transfer air from the air cleaner outlets to the compressor intakes on the turbochargers. Each clean air duct incorporates a convolute section to accommodate relative movement between the engine mounted turbochargers and the body mounted air cleaners. The left clean air duct also incorporates a quarter wave resonator to reduce air induction noise.

If bi-turbo variant is fitted the left clean air duct also incorporates connection for the bypass hose from the charge air intake valve.
For additional information, refer to: Turbocharger (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Description and Operation).

Brackets on the clean air ducts attach the ducts to the A/C compressor (left air duct) and the generator mounting bracket (right clean air duct, if equipped).

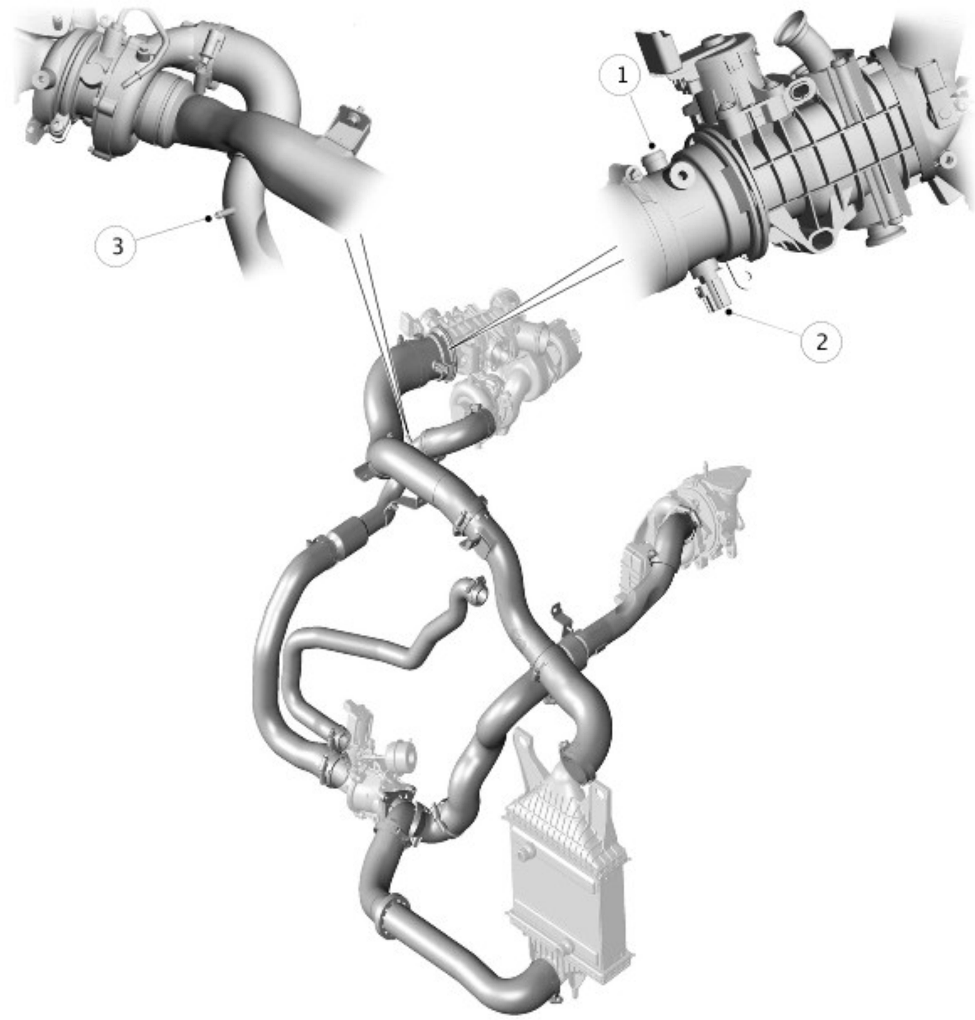
TURBOCHARGERS

The turbochargers are attached to the exhaust manifolds. A variable geometry turbine (VGT) turbocharger, designated as the primary turbocharger, is located on the left side of the engine. On vehicles with bi-turbo variant a fixed vane turbocharger, designated as the secondary turbocharger, is installed on the right side of the engine.

For additional information, refer to: Turbocharger (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Description and Operation).

CHARGE AIR DUCTS

Charge Air Ducts – Twin-turbo variant



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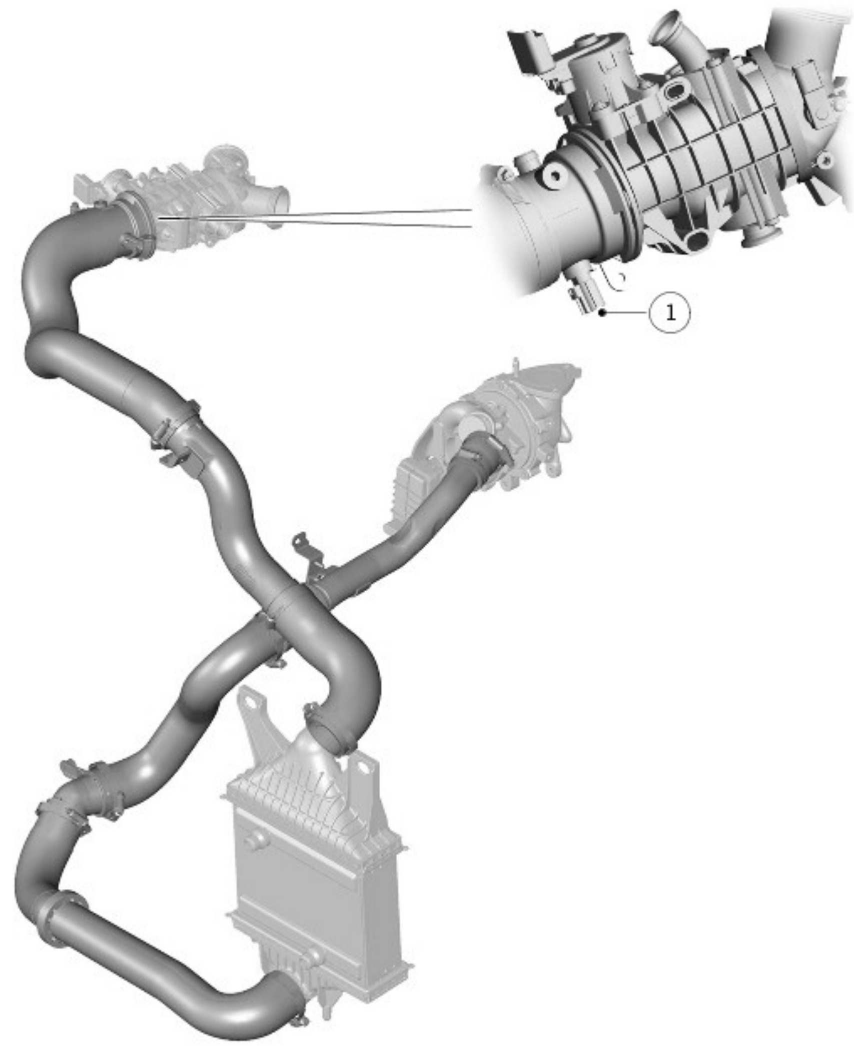
ITEM	DESCRIPTION
1	Secondary turbocharger air tube connection
2	Charge air temperature sensor

3	Secondary turbocharger charge air pressure connection

Charge Air Ducts – Mono-turbo variant



ITEM	DESCRIPTION
1	Secondary turbocharger air tube connection
2	Charge air temperature sensor

The charge air ducts interconnect the turbocharger compressor outlets, charge air intake valve (if bi-turbo variant is equipped), charge air cooler and electric throttle.

The charge air duct connection with the electric throttle incorporates connections for:

- An air tube connected to the secondary turbocharger on bi-turbo vehicles.
  For additional information, refer to: Turbocharger (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Description and Operation).
- A charge air temperature sensor connected to the ECM.
  For additional information, refer to: Electronic Engine Controls (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation).

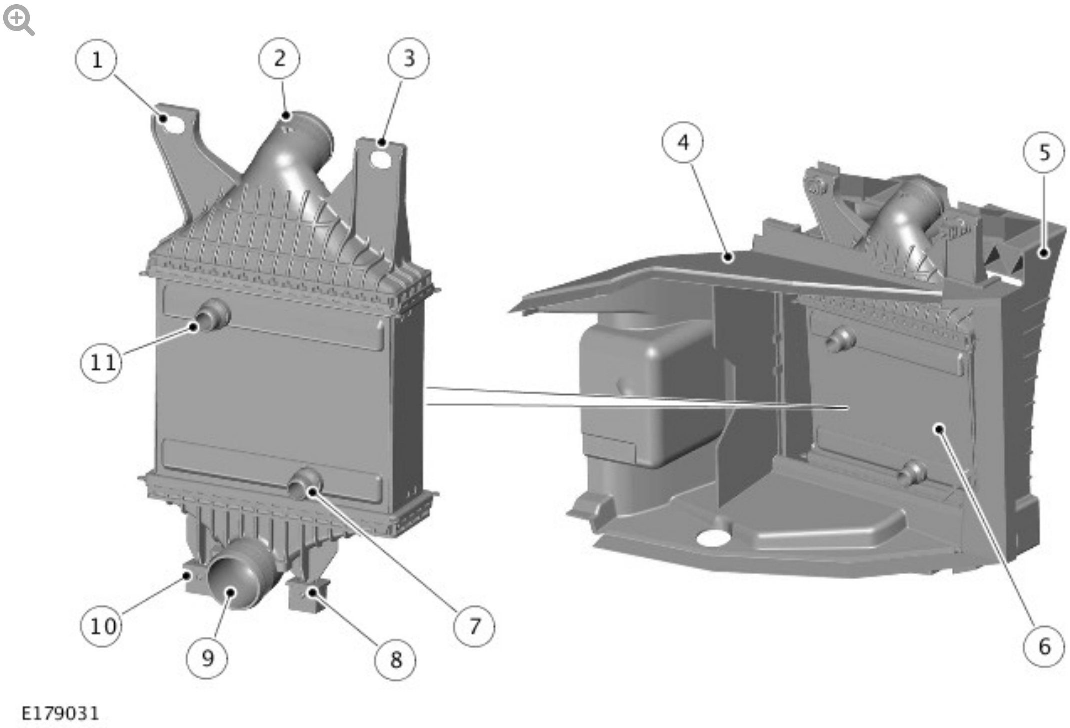
The charge air duct connected to the outlet of the secondary turbocharger incorporates a connection for the charge air pressure sensor.
For additional information, refer to: Electronic Engine Controls (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation).

CHARGE AIR INTAKE VALVE – ONLY FOR BI-TURBO VARIANT

The charge air intake valve is attached to a bracket on the front subframe and the cooling pack protector.

The charge air intake valve is used by the ECM to change turbocharger operation between mono and bi-turbo modes. The valve controls the flow of air from the compressor of the secondary turbocharger to the clean air duct of the primary turbocharger and to the charge air cooler.
For additional information, refer to: Turbocharger (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Description and Operation).

CHARGE AIR COOLERS

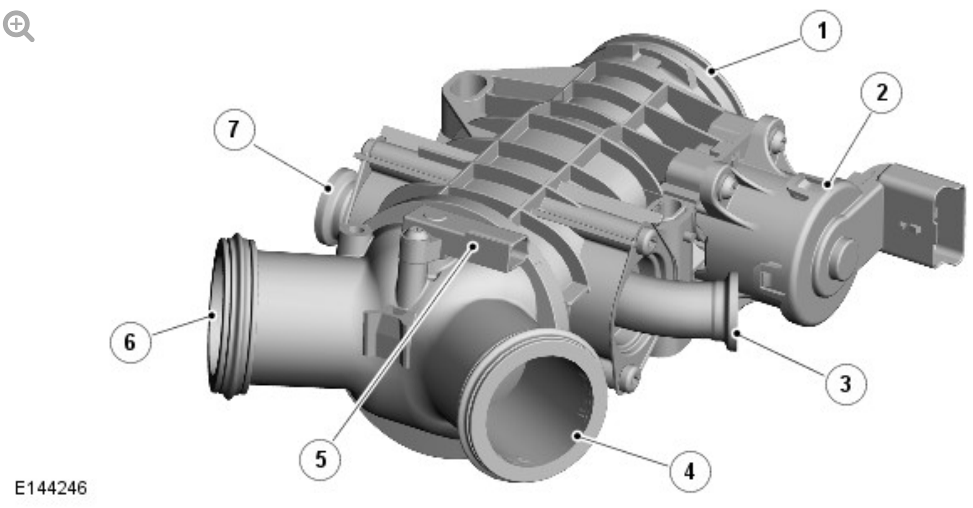


ITEM	DESCRIPTION
1	Isolator bush
2	Air intake
3	Isolator bush
4	Front air duct
5	Rear air duct
6	Charge air cooler
7	Cooling water connection
8	Isolator
9	Air outlet
10	Isolator
11	Cooling water connection

The charge air cooler is a water cooled charge air cooler that reduce the intake air temperature.

A charge air cooler is installed on the left side of the cooling pack, behind the left outer air intake in the front bumper. The charge air cooler is installed in air duct attached to the front bumper armature and the front end carrier. The lower end tank of the charge air cooler is located in the air ducts by two spigots fitted with isolator bushes. The upper end tank of each charge air cooler is attached to the front end carrier and the air ducts with two isolator bushes, nuts and bolts.

ELECTRIC THROTTLE



ITEM	DESCRIPTION
1	Air intake
2	Electric motor
3	Right exhaust gas recirculation pipe connection
4	Right intake manifold connection
5	Manifold absolute pressure sensor
6	Left intake manifold connection
7	Left exhaust gas recirculation pipe connection

The electric throttle regulates the air flow from the charge air cooler to the intake manifolds.

The electric throttle is installed at the front of the engine, between the cylinder heads. Two outlets on the electric throttle locate in the two intake manifolds. A stud bolt attaches the electric throttle to the right cylinder head.

The throttle plate is operated by a direct current (DC) electric motor attached to the throttle body. The motor is controlled by the ECM and is constantly adjusted in response to driver inputs with the throttle pedal to precisely control the amount of air allowed into the intake manifolds.
For additional information, refer to: Electronic Engine Controls (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation).

Pipe connections on both sides of the throttle body allow for the attachment of the outlet pipes from the Exhaust Gas Recirculation (EGR) valves. The pipe attachments in the throttle body are specifically designed to mix the recirculated exhaust gas with the intake air and provide an even distribution to each side of the engine.

A manifold absolute pressure (MAP) sensor is located in the top of the throttle body where the air flow splits for the two intake manifolds.
For additional information, refer to: Electronic Engine Controls (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation).

INTAKE MANIFOLDS

The intake manifolds are an integral part of the camshaft covers. Each intake manifold is connected to the electric throttle by a push fit, sealed connection.
For additional information, refer to: Engine (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).